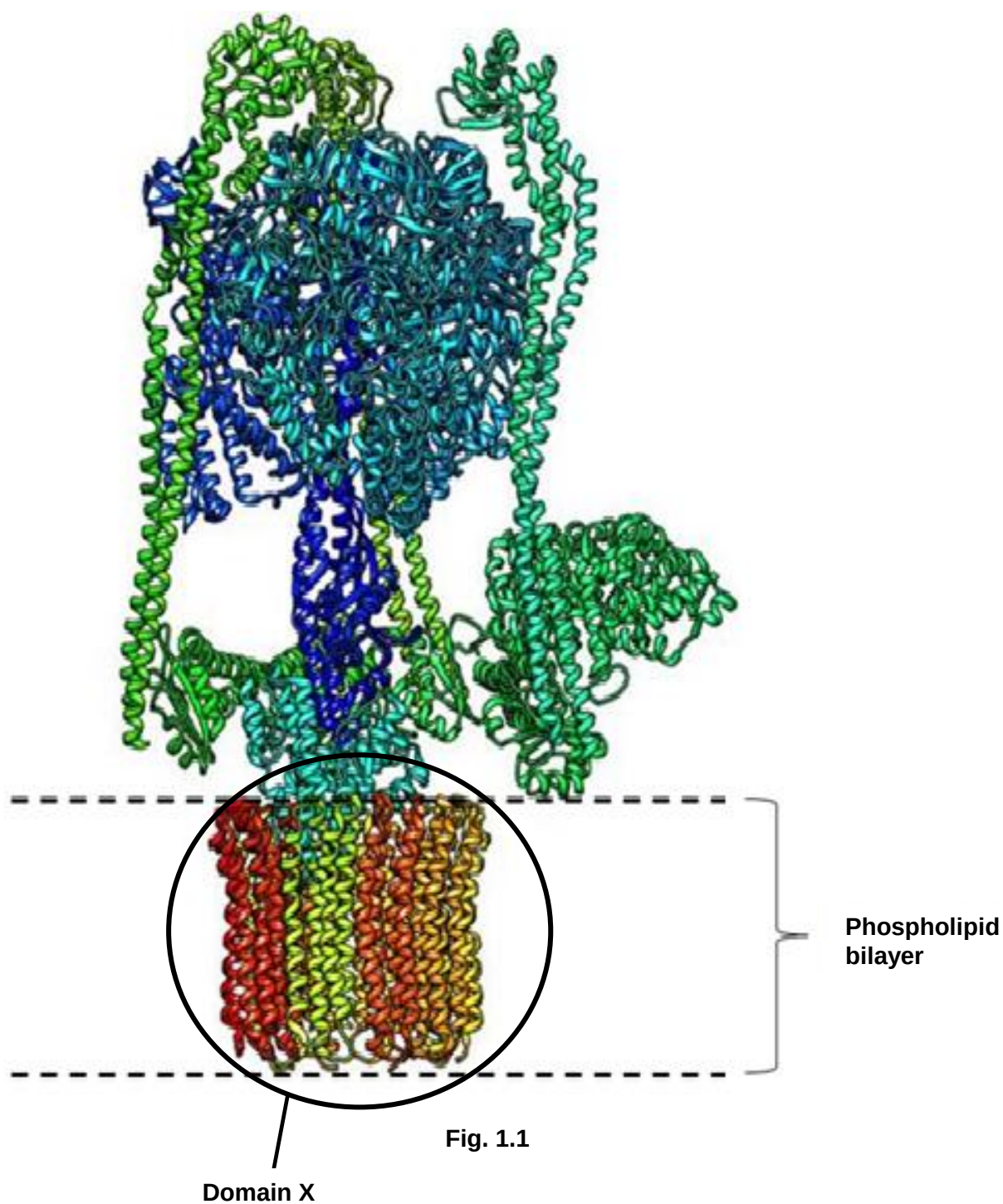


SECTION A

Answer **all** questions.

Question 1

Transport proteins are embedded within the membranes of various organelles. **Fig. 1.1** shows the structure of one such transport protein. The protein is known to contain multiple polypeptide chains.



(a) With reference to **Fig. 1.1**, describe how the structure of Domain X is maintained [2]

1. **primary structure/polypeptide chain** folds into **secondary structures** which are maintained by **hydrogen bonds** formed between **C=O and NH groups**;
2. **Several α -helices** arranged such that they are **parallel** to each other;
3. **further folding** into a unique **3D conformation** maintained by **hydrogen bonds, ionic bonds, disulfide bonds** and **hydrophobic interactions** between **R-groups** of amino acid residues;

Max 2

A researcher intends to study the transport of substances carried out by the transport protein. The researcher begins by creating artificial vesicles containing only the transport protein embedded within the membrane. The artificial vesicles were then placed in 5 different solutions, each containing a different substance with a concentration of 10 arbitrary units.

The concentrations of the substances in the artificial vesicles were measured before the experiment and after 15 minutes of the experiment. The results obtained by the researcher are shown in **Table 1.2**.

Solution	Concentration of substances in the artificial vesicles before experiment	Concentration of substances in the artificial vesicles after 15 minutes of the experiment
1	0 AU of Na ⁺ ions	0 AUs of Na ⁺ ions
2	0 AUs of Cl ⁻ ions	0 AUs of Cl ⁻ ions
3	1 AUs of H ⁺ ions	20 AUs of H ⁺ ions
4	0 AU of protein insulin	0 AU of protein insulin
5	0 AU of amino acid glutamate	0 AUs of acidic amino acid glutamate

Table 1.2

(b) With reference to **Table 1.2**, explain the role of the transport protein in transporting substances across membranes. [4]

1. The concentration of H⁺ ions in the artificial vesicles after 15 minutes is **20AU**, which is **higher** than the initial concentration of **10AU** in the solution;
2. **Active transport of substances** is carried out by **carrier protein/ protein pump**;
3. Which transports only a **specific** substance, which is **H⁺ ions** due to presence of **specific binding sites for H⁺ ions**;
4. Ref. to **against concentration gradient** from **outside** of artificial vesicles to **inside** artificial vesicles, using **energy from ATP hydrolysis**;

Another Protein A is a receptor protein. However, unlike the transport protein shown in **Fig. 1.1**, it is made up of only a single polypeptide and is embedded within the cell surface membrane.

(c) Outline the route taken by Protein A after it is synthesized by the ribosome on the rough endoplasmic reticulum. [3]

1. Protein A inserted/embedded (reject fuse/combine) into the membrane of the rough endoplasmic reticulum;
2. Vesicles with protein embedded in membrane bud off and move to Golgi apparatus;
3. Vesicles bud off from golgi apparatus and fuse with the cell surface membrane;

[Total: 9]

Question 2

Fig. 2.1 shows cultured fibroblasts of the Indian barking deer, *Muntiacus muntjac*, undergoing different stages of the mitotic cell cycle, **J** to **Q**.

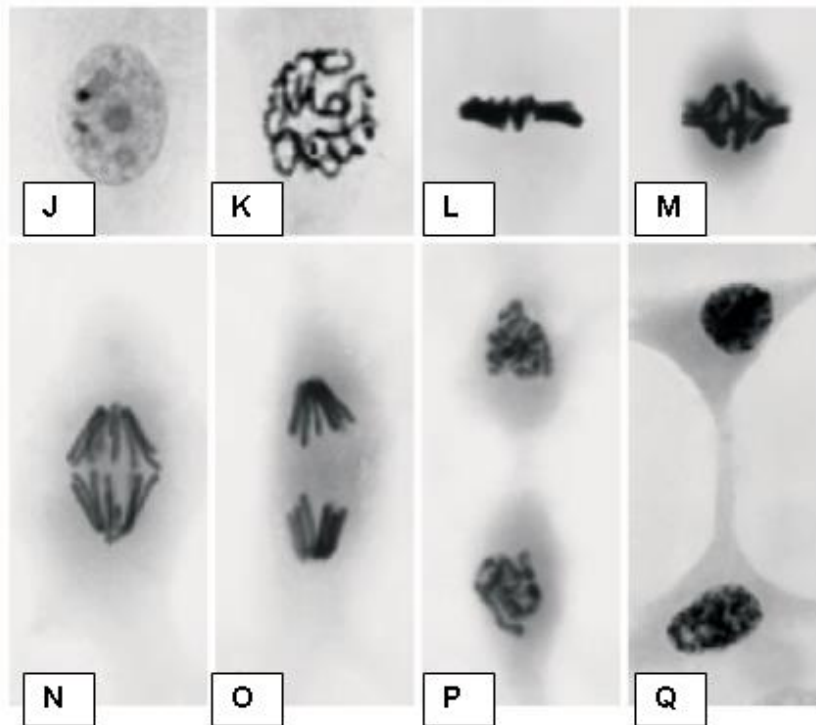


Fig. 2.1

(a) With reference to **Fig. 2.1**, identify stages **K**, **L**, **N** and **Q**.

[2]

K: prophase (reject interphase) **L:** metaphase

N: anaphase

Q: cytokinesis [0.5 mark each]

(b) Describe the events that occur in stage P. [2]

1. Individual chromosomes arrive at opposite poles of the cell;
2. Chromosome uncoil/decondense and become thread-like CHROMATIN;
3. A nuclear envelope reforms round the chromosomes at each pole;
4. Nucleoli reappear/reform;
5. Spindle fibres disassemble

Max 2

(c) Outline the roles of centromeres in mitosis. [3]

1. allow sister chromatids to adhere/join to each other;
2. allow spindle fibres to attach at kinetochores on centromere of sister chromatids;
3. when centromeres divide, this leads to equal separation of sister chromatids to opposite poles during anaphase.

(d) Fig. 2.2 shows changes in DNA content at different stages of the animal life cycle.

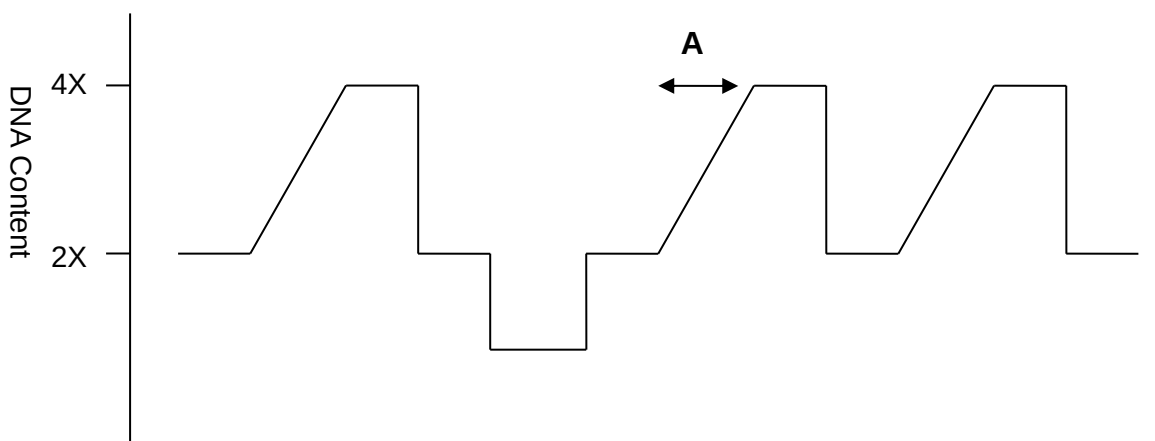


Fig. 2.2

With reference to Fig. 2.2,

(i) identify and describe Stage A. [2]

1. DNA replication during Interphase;
2. Doubling of DNA content from 2X to 4X;
3. Each of 2 parental DNA strands act as templates for synthesis of complementary daughter strands;
4. DNA polymerase elongates new strand formed in 5' to 3' direction

Max 2

- (ii) state the significance of Stage A. [1]

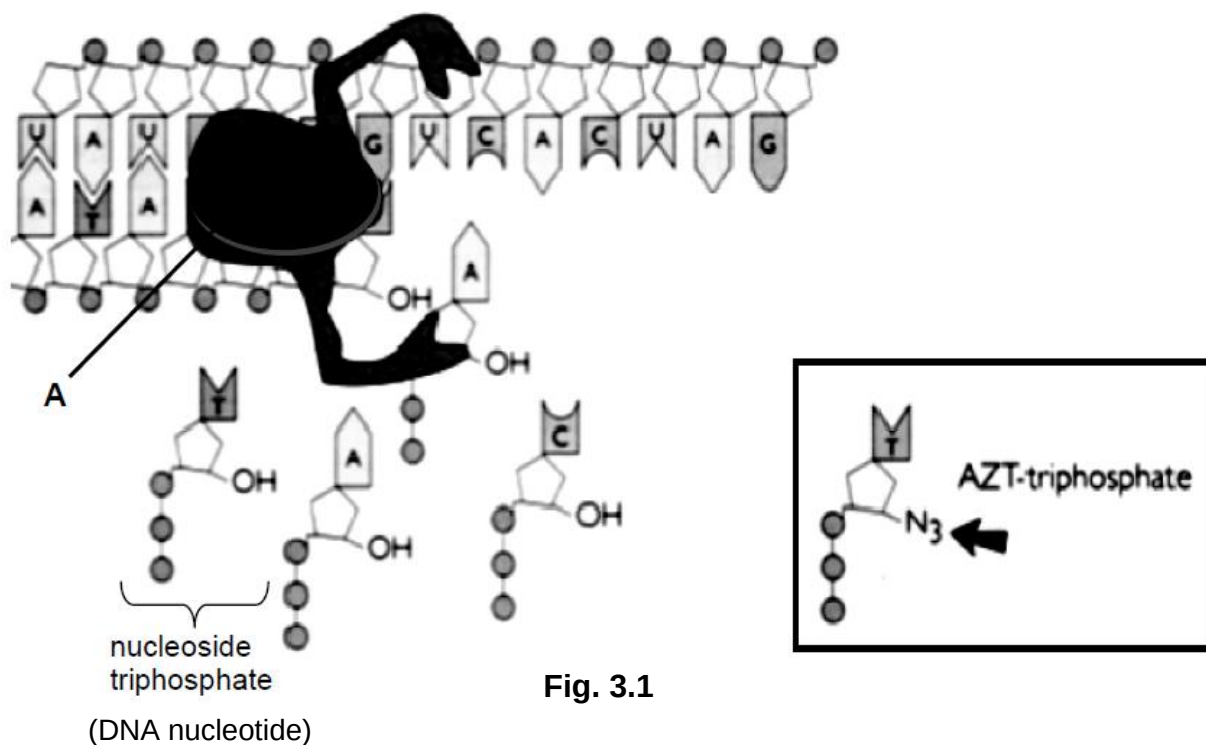
1. to produce genetically identical daughter cells

[Total: 10]

Question 3

- (a) The Human Immunodeficiency Virus (HIV) causes Acquired Immune Deficiency Syndrome (AIDS).

Fig. 3.1 shows the molecular process in an infected host cell that has the anti-viral drug AZT-triphosphate introduced.



- (i) Name enzyme **A** and describe the process it catalyses. [2]

1. Reverse transcriptase;
 2. Synthesis of complementary DNA using an RNA template through formation of phosphodiester bonds between deoxyribonucleotides/ DNA nucleotides

- (ii) Explain the action of AZT-triphosphate as an anti-viral drug. [3]

1. AZT-triphosphate is similar in conformation to thymidine triphosphate / can be recognised by the enzyme reverse transcriptase / AZT-triphosphate contains a thymine base;

2. Can form complementary base pairing with adenine on RNA strand/ Allowing it to be incorporated into the newly synthesised DNA strand;
3. it stops strand elongation as it lacks the 3' OH group for the formation of the phosphodiester bond with the next nucleotide hence terminating the elongation of the DNA strand

NOTE: this drug is inhibiting the replication of viral genome, thus an anti-viral drug

- (iii) With reference to the structure of A, suggest one way in which AZT-triphosphate loses its effectiveness as an anti-viral drug. [2]

1. The 3D conformation of the reverse transcriptase is altered through mutation;
2. such that it can distinguish between AZT and Thymidine and hence not incorporating AZT

- (b) Describe how the reproductive cycles of lambda phage differs from HIV. [2]

Max 2

Feature of comparison	Lambda phage	HIV
Host cell	<u>Bacteria</u>	<u>Human CD4 T helper cells</u>
Attachment / Adsorption	Uses <u>tail fibres</u> bind to <u>specific receptor</u> sites on <u>bacteria outer cell membrane</u> ;	<u>Glycoproteins 120 (gp120)</u> on <u>viral envelope</u> recognize / bind to <u>specific CD4 receptor</u> molecules on cell surface membrane of CD4 helper T cell;
Penetration / Entry	Phage <u>injects</u> only its <u>double-stranded (d/s) DNA/genome</u> ;	<u>HIV releases its nucleocapsid</u> (single-stranded RNA and reverse transcriptase) into cytoplasm;
Fate of viral nucleic acids	<u>No reverse transcription</u> as phage's genetic material is <u>dsDNA/ ds DNA</u> is either <u>immediately used as template</u> for synthesis of viral proteins and nucleic acids (lytic cycle)	Its single stranded <u>RNA</u> is converted to <u>double-stranded DNA</u> , using <u>viral reverse transcriptase</u> ;
	Can enter lytic phase or <u>integrated into host chromosome to form prophage</u> (lysogenic phase);	<u>Viral genome incorporated into host cell chromosomes (provirus)</u> ;
	<u>Prophage is excised</u> from host cell's chromosome upon <u>spontaneous induction</u> ;	<u>Provirus is not excised</u> as viral mRNA is transcribed from viral DNA together with host cell's genes;

Synthesis of viral nucleic acids / Replication of viral genome	Its <u>double-stranded DNA</u> (genome) is synthesized <u>directly</u> using <u>host cell's</u> enzymes such as <u>DNA polymerase</u> ;	Its <u>ssRNA</u> (genome) is synthesized <u>indirectly</u> using its <u>reverse transcriptase</u> and <u>host cell's</u> enzymes such as <u>RNA polymerase</u> ;
Site of synthesis of viral nucleic acids / replication of viral genome	<u>Cytoplasm</u> of bacterium;	<u>Cytoplasm</u> (reverse transcription) and <u>nucleus</u> (transcription) of human cell;
Integration of viral glycoproteins into cell membrane	No integration of viral glycoproteins into host's cell membrane;	Integration of <u>viral glycoproteins</u> (envelope gp120 and gp41) into <u>host's</u> cell surface membrane;
Release / Exit	Host cell is <u>lysed</u> to release the new viruses during lytic cycle	The new viruses <u>bud off</u> from <u>host cell's</u> cell surface membrane, studded with <u>viral glycoproteins</u>

[Total: 9]

Question 4

- (a) Fig. 4.1 illustrates the structure of a telomere. The looped segment of the telomere shown is the T loop.

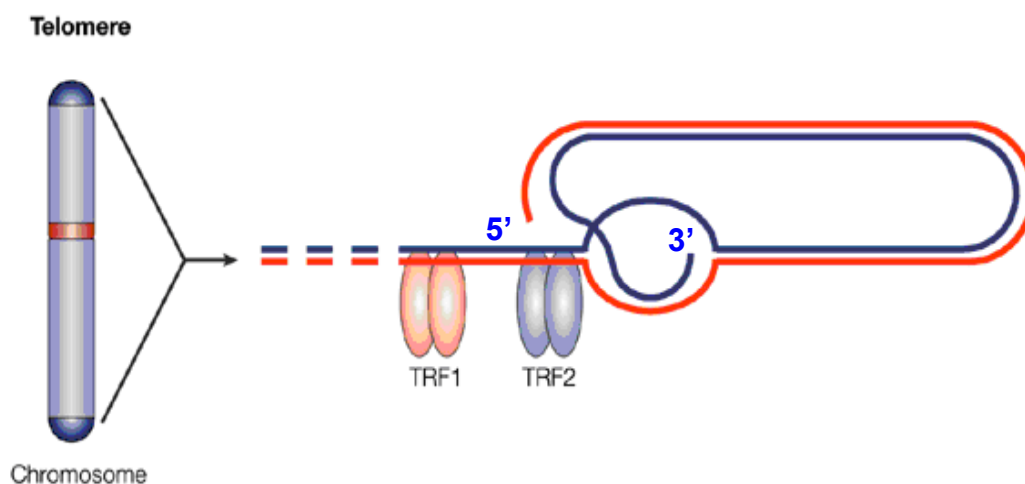


Fig. 4.1

- (i) Label on the diagram, the 3' and 5' ends of the DNA. [1]
- (ii) Explain the phenomenon that enabled you to distinguish the 3' and 5' ends of the DNA in part (a) (i). [3]

1. **3' end is always longer** than 5' end of a **linear DNA** molecule due to the **end replication problem**;
2. **DNA polymerase** can only add **DNA nucleotides** to the free **3' OH** end of a **pre-existing polynucleotide chain**;
3. When the last **RNA primer** at **5' end** of DNA is **removed**, **DNA is not replaced/ gap is not filled** in as there is no **available 3' OH group**, thus, resulting in a **shorter 5' end**.

(b) Describe the role of telomeres.

[2]

1. Serves as **buffer DNA** to protect the **organism's genes** from being eroded after successive rounds of DNA replication;
2. **Maintain stability of chromosome** by **preventing single-stranded ends** of chromosomes from **fusing** via formation of **T and D loops**;
3. Prevent **chromosomal ends** from **activating** the cell's system for **monitoring DNA damage**;
4. Plays a **protective function** by **preventing chromosomes ends** from **degradation** by cellular **nucleases** via the formation of **telomeric cap**;
5. **Shortens** with every cell division/ DNA replication once **critical length** of telomere is reached, then cells undergo senescence and **apoptosis** OR
6. Indirectly **prevents cancer development** by **preventing the accumulation of mutations** as old cells with shortened telomeres undergo **apoptosis**

Max 2

Fig. 4.2 shows the growth pattern of bacteria cells in a medium containing a mixture of glucose and lactose. Observations of the concentrations of the two sugars in the medium show that glucose falls to very low levels after the start of the experiment, but lactose remains relatively high and only starts to decrease near the end of the experiment (not shown in **Fig. 4.2**).

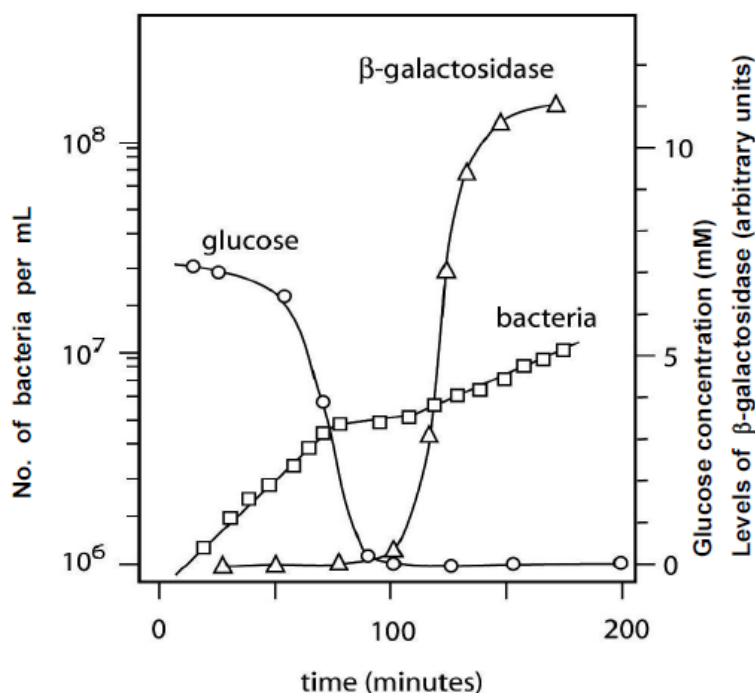


Fig. 4.2

From the experiment, it was observed that the growth rate pattern of bacteria cells showed a rapid increase in number, constant numbers in the middle of the experiment followed by a slower increase towards the end.

- (c) Suggest why the number of bacteria cells remained constant during the middle of the experiment. [1]

Number of bacteria cells remained constant remain constant because

1. Glucose is used up AND bacteria cells have not produced sufficient β -galactosidase for lactose metabolism;
2. Time is needed for the expression of the lac operon/ production of β -galactosidase before lactose is hydrolysed/metabolized;
3. Time is needed for lactose to be transported into the cell and converted into allolactose.

Max 1

- (d) With reference to **Fig. 4.2**, explain why β -galactosidase levels only increased after 100 minutes. [3]

1. Glucose concentration reaches 0mM at 100 min which leads to an increase in cAMP concentration; [Describe]
2. cAMP binds to allosteric site on catabolite activator protein (CAP), this changes conformation of CAP and CAP is now in active form;
3. CAP binds to CAP binding site on DNA upstream of promoter and this bends the DNA;
4. making it easier for RNA polymerase to bind to the promoter and transcription of genes in lac operon occurs to produce β -galactosidase.

Max 3

In prokaryotes, genes are often organized into operons.

- (e) Using named examples, discuss the differences between inducible and repressible operons. [2]

Max 2

	Inducible operon	Repressible operon
Named example	<u>Lac operon</u>	<u>Trp operon</u>
Nature of metabolic pathway	Function in <u>catabolic</u> pathways	Function in <u>anabolic</u> pathways
Differences in regulation	<u>Usually [not expressed/ turned off], but can be [stimulated/ induced/ turned on]</u>	<u>Usually [turned on/ expressed], can be [inhibited/ repressed/ turned off]</u>
Nature of repressor protein	Synthesizes <u>active repressor</u>	Synthesizes <u>inactive repressor</u>

<u>Effect on repressor protein</u> & operon when <u>lactose/inducer; corepressor/ tryptophan is present</u> OR	<u>[Allolactose/ inducer] binds repressor</u> , repressor becomes <u>inactive</u> , [transcription turned on/ operon expressed] OR	<u>[Co-repressor/ tryptophan] binds repressor</u> , repressor becomes <u>active</u> , [transcription turned off/ operon not expressed] OR
<u>Effect on repressor protein</u> & operon when <u>lactose/inducer; corepressor/ tryptophan is absent</u>	<u>Active repressor binds operator</u> , [transcription <u>turned off</u> / operon <u>not expressed</u>]	<u>Inactive repressor cannot bind operator</u> , [transcription <u>turned on</u> / operon <u>expressed</u>]

Question 5

A group of genetics students was investigating the inheritance of body colour and wing shape in a new species of fly. Pure-breeding flies with wild-type bodies and vestigial wings were crossed with pure-breeding flies with black bodies and wild-type wings.

All the F₁ offspring had wild-type bodies and wild-type wings.

Flies from the F₁ were crossed with black bodied, vestigial-winged males. 2876 offspring flies were produced as shown below:

wild-type body, vestigial wings	1277
wild-type body, wild-type wings	143
black body, vestigial wings	152
black body, wild type wings	1304

- (a) Based on the information and results given above, describe the inheritance of body colour and wing shape in this species of fly. [2]

- The **two genes** are located on the **same chromosome** / **linked genes**;
- Wild-type body** is **dominant** to black body and **wild-type wings** is **dominant** vestigial wing; **[accept recessive traits]**

- (b) Draw a genetic diagram to explain the observed results of the cross between F_1 flies and black bodied, vestigial-winged flies.

Use the following symbols to represent the different alleles involved:

A/a for gene locus controlling body colour,

B/b for gene locus controlling wing shape.

[4]

Phenotypes:	wild-type body, wild-type wings x black body, vestigial wings				1M										
Genotypes:	<u>A</u> <u>b</u> <u>a</u> <u>B</u> AaBb X <u>a</u> <u>b</u> <u>a</u> <u>b</u> aabb														
Gametes:	<u>A</u> <u>b</u> <u>Parental gametes</u> <u>a</u> <u>B</u> <u>Recombinant gametes</u> <u>A</u> <u>B</u> <u>Recombinant gametes</u> <u>a</u> <u>b</u> <u>Parental gametes</u> X <u>a</u> <u>b</u>				1M										
Offspring genotype and phenotype:	<table><tr><td></td><td> <u>A</u> <u>b</u> </td><td> <u>a</u> <u>B</u> </td><td> <u>A</u> <u>B</u> </td><td> <u>a</u> <u>b</u> </td></tr><tr><td> <u>a</u> <u>b</u> </td><td> <u>A</u> <u>b</u> <u>a</u> <u>b</u> <u>Wild-type body, vestigial wings</u> </td><td> <u>a</u> <u>B</u> <u>a</u> <u>b</u> <u>black body, wild-type wings</u> </td><td> <u>A</u> <u>B</u> <u>a</u> <u>b</u> <u>Wild-type body, Wild-type wings</u> </td><td> <u>a</u> <u>b</u> <u>a</u> <u>b</u> <u>black body, vestigial wings</u> </td></tr></table>					<u>A</u> <u>b</u>	<u>a</u> <u>B</u>	<u>A</u> <u>B</u>	<u>a</u> <u>b</u>	<u>a</u> <u>b</u>	<u>A</u> <u>b</u> <u>a</u> <u>b</u> <u>Wild-type body, vestigial wings</u>	<u>a</u> <u>B</u> <u>a</u> <u>b</u> <u>black body, wild-type wings</u>	<u>A</u> <u>B</u> <u>a</u> <u>b</u> <u>Wild-type body, Wild-type wings</u>	<u>a</u> <u>b</u> <u>a</u> <u>b</u> <u>black body, vestigial wings</u>	1M
	<u>A</u> <u>b</u>	<u>a</u> <u>B</u>	<u>A</u> <u>B</u>	<u>a</u> <u>b</u>											
<u>a</u> <u>b</u>	<u>A</u> <u>b</u> <u>a</u> <u>b</u> <u>Wild-type body, vestigial wings</u>	<u>a</u> <u>B</u> <u>a</u> <u>b</u> <u>black body, wild-type wings</u>	<u>A</u> <u>B</u> <u>a</u> <u>b</u> <u>Wild-type body, Wild-type wings</u>	<u>a</u> <u>b</u> <u>a</u> <u>b</u> <u>black body, vestigial wings</u>											
Offspring phenotypic ratio:	wild-type body, vestigial wings : black body, wild type wings : wild-type body, wild-type wings : black body, vestigial wings 1277 : 1304 : 143 : 152				1M										

- (c) The genetics students then investigated the inheritance of body length of the fly. Pure-breeding flies with adult body length of more than 3.8 mm were crossed with pure-breeding flies with adult body length of less than 2.4 mm.

It was observed that the F_1 offspring showed a range of adult body lengths with most flies having a body length of 3.0 mm. When the F_1 offspring were crossed with each other, the F_2 offspring were observed to have a wider range of adult body length than the F_1 offspring.

- (i) State the term used to describe the range of body lengths in the F_1 and F_2 offspring. [1]

1. Continuous variation

- (ii) Explain why a wide range of body lengths were observed in the F_2 offspring compared to the F_1 offspring. [3]

1. Body length is the **combined/additive** effect of **many genes/polygenes**;
2. Each gene having a **little overall effect** on body length;
3. Wider range of body lengths in F_2 offspring is due to **F_2 generation having a wider range of genotypes/not genetically identical** whereas **F_1 generation individuals have the same genotype/genetically identical**;
4. Wide range of genotypes in F_2 due to independent assortment of homologous chromosomes / crossing over & chiasmata formation at prophase I of meiosis occurring during **formation of F_1 gametes**, resulting in the formation of **recombinant gametes** with new combination of alleles;

OR

5. due to random mating and random fusion of gametes during fertilisation;

max 3

[Total: 10]

Question 6

- (a) **Fig. 6.1** shows the cell signalling pathway by hormone Epidermal Growth Factor (EGF).

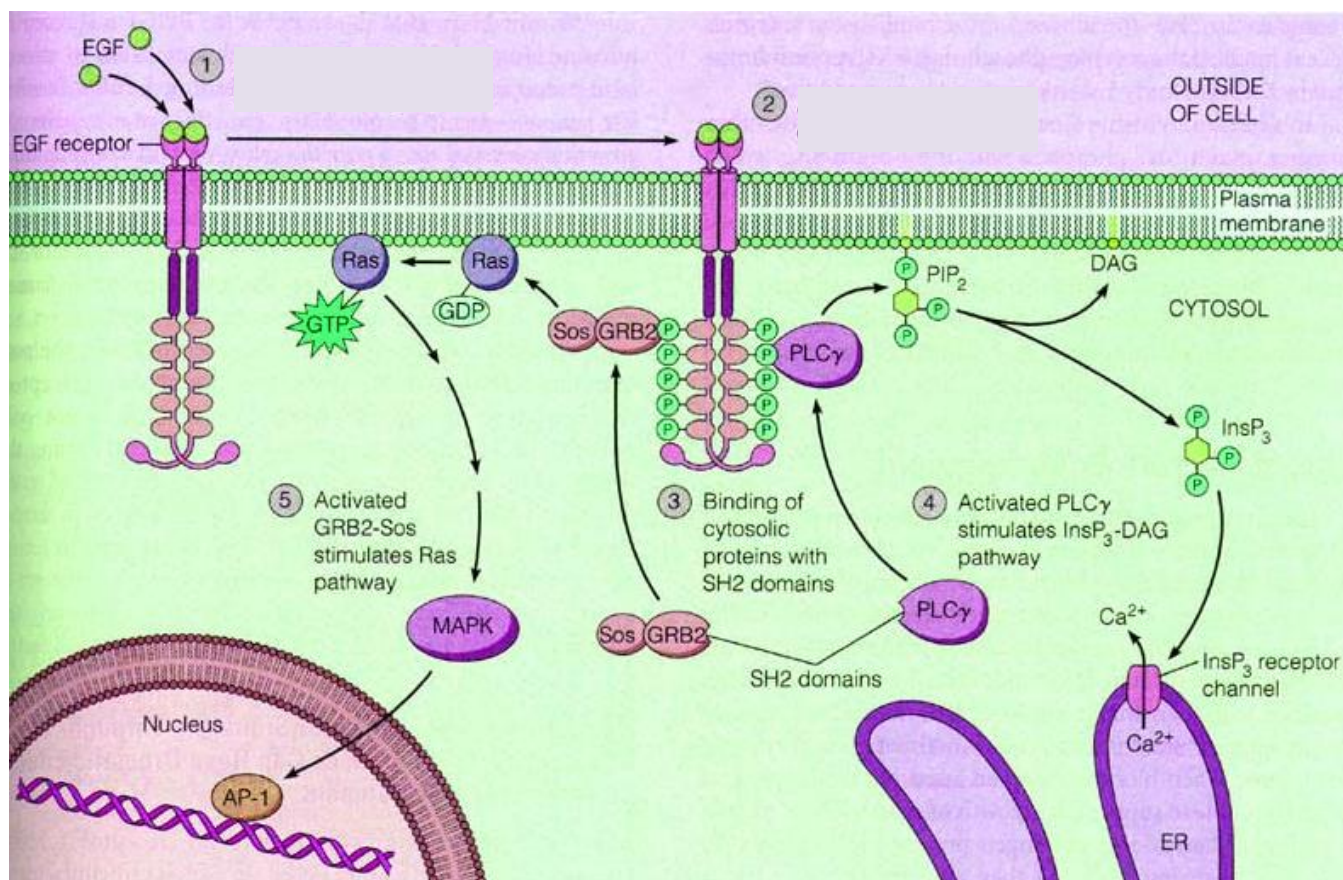


Fig. 6.1

- (i) With reference to **Fig. 6.1**, describe the events occurring in steps 1 and 2. [3]

- Step 1:** Binding of ligands EGF to extracellular ligand-binding site of specific EGF receptors causes the two polypeptide chains to dimerise;
- Step 2:** The intracellular kinase region on each chain is activated to autophosphorylate;
- By transferring phosphate groups from ATP molecules to tyrosine residues on the opposite/ other chain

- (ii) When Ras protein is activated, it could result in large numbers of activated MAP Kinase (MAPK). Suggest what could happen during stage 5 that would lead to such an outcome. [2]

- Signal amplification has occurred whereas number of molecules activated at each step is greater than the preceding step;
- This is achieved via phosphorylation cascade involving a series of kinases each phosphorylates and activates many more downstream kinases, finally leading to large amounts of MAPK being activated.

- (iii) The receptor for EGF can be activated by a deletion that removes part of its extracellular domain. These mutant receptors are able to form active dimers even in the absence of EGF, and thus they produce a stimulatory signal inappropriately. Besides mutation to the gene that codes for the extracellular domains of the receptors, 30% of human tumors have a ras mutation.

With reference to **Fig. 6.1**, suggest and explain how a missense mutation in the Ras gene may bring about cancer. [3]

1. Missense mutations coding for different amino acids may result in a change in 3D conformation of the Ras protein;
2. Causing **Ras protein** to be hyperactive - The **GTPase activity of Ras** may be destroyed (due to change in the shape of active site), rendering it unable to hydrolyse GTP back to GDP/ remains GTP bound;
3. **Hyperactive Ras** constitutively/continuously activate relay proteins/ kinases (even in the absence of the extracellular signal molecule/ growth factor), thus,
4. stimulates uncontrolled cell proliferation and cell division, leading to tumor formation.

[Total: 8]

Question 7

- (a) Fig. 7.1 shows the changes in membrane potential at one point on an axon of a nerve cell.

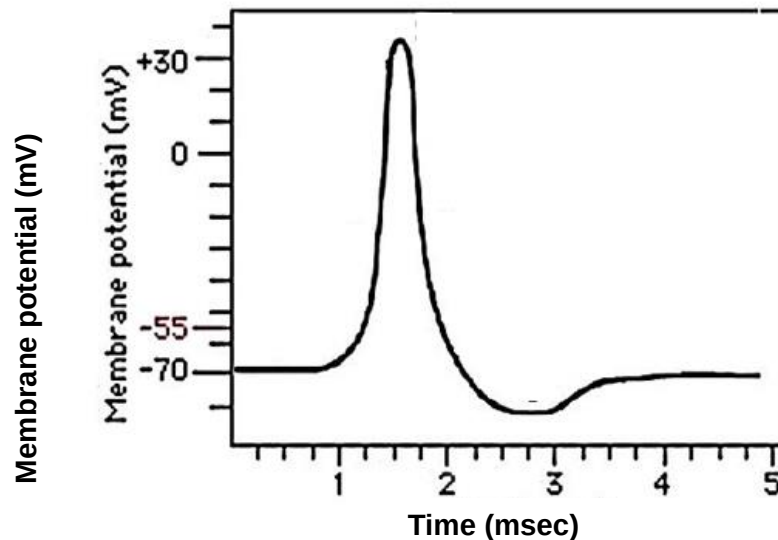


Fig. 7.1

With reference to Fig. 7.1, explain the shape of the graph

- (i) between 1.0 msec and 1.5 msec [2]
- between 1.0 msec to 1.5 msec, depolarisation occurs, whereby membrane potential increases from -70mV to +35mV; (ACCEPTABLE range +33mV to +38mV)
 - voltage-gated Na⁺ ion channels (in the membrane) open, allowing diffusion/influx of Na⁺ ions into neurone.
- (ii) between 2.5 msec and 3.0 msec [2]
- between 1.5msec to 2msec, hyperpolarisation occurs, whereby membrane potential decreases till -80 mV; (ACCEPTABLE range -80mV to -90mV);
 - this is due to the slow/delayed closure of voltage-gated K⁺ channels, resulting in excessive efflux of/ diffusion out of K⁺ ions, thus membrane potential becomes more negative than resting potential
- (b) Suggest why transmission of impulses along a myelinated axon uses less energy than an unmyelinated axon of the same diameter. [2]
- In myelinated axons, action potentials are formed only at the nodes of ranvier & thus active transport of ions occurs at nodes of Ranvier only;
 - less Na⁺/K⁺ pumps are required to restore resting membrane potential hence less energy in form of ATP required

- (c) A new painkiller that is much more effective than morphine became available in the United Kingdom in 2006. It was extracted from the venom of the Magician's Cone Snail, *Conus magus*.

C. magus is a native of coral reefs in the Pacific Ocean. It injects the venom into fish, which are paralysed and swallowed whole. Scientists have discovered that the venom blocks calcium channels at synapses between neurones.

Explain how the blocking of calcium channels can result in both reduction of pain in humans and paralysis in fish. [3]

1. no uptake of calcium ions into presynaptic membrane / terminal/ synaptic knob;
2. no migration of synaptic vesicles towards presynaptic membrane / fusion of synaptic vesicles with presynaptic membrane;
3. no exocytosis / no release of neurotransmitter / acetylcholine into the synaptic cleft;
4. no diffusion of neurotransmitter across synaptic cleft;
5. no binding of neurotransmitter to RECEPTORS on postsynaptic membrane AND no depolarisation of postsynaptic membrane/ no EPSP (excitatory postsynaptic potential generated)

- (d) Diabetes mellitus is often due to the destruction of the beta-cells in the islets of Langerhans in the pancreas. **Fig. 7.2** shows the changes in blood glucose concentration over a day in which an untreated diabetic person and a non-diabetic person take the same meals and exercise. Blood glucose was measured in both individuals every hour. At meal times, the measurements were made just before the meal.

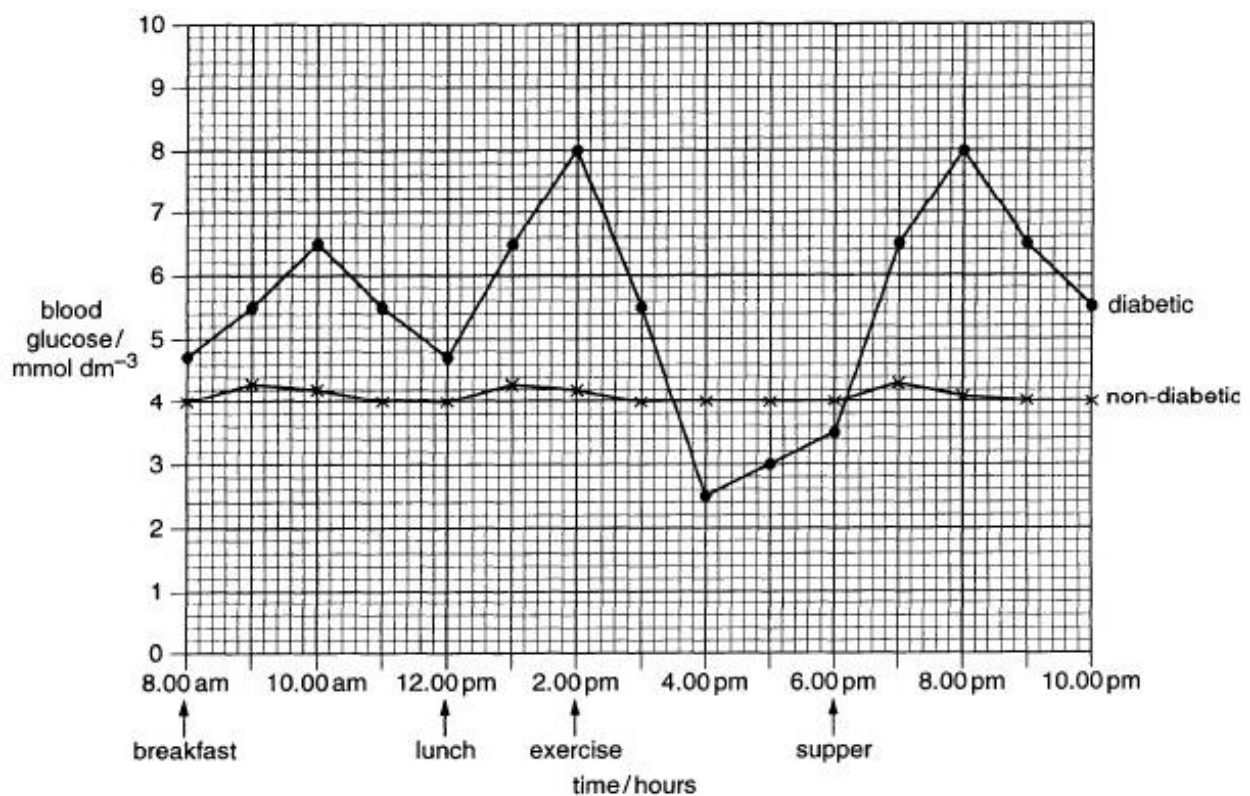


Fig . 7.2

Explain why, following exercise at 2.00 pm,

- (i) the blood glucose in the diabetic person dropped; [1]
1. exercise increases need for ATP / respiration, thus muscles take up glucose from the blood;
 2. in diabetic little or no glycogen stored, to replace blood glucose taken up
- (ii) the blood glucose in the non-diabetic person stayed the same. [2]
1. ref to action of glucagon secreted by α -cells in response to fall in blood glucose levels;
 2. glucagon stimulates liver cells to break down glycogen to form glucose;
 3. which replace blood glucose taken up / maintain blood glucose concentration at the norm levels

[Total: 12]

Question 8

A population of a diploid species inhabiting remote island X was tracked for many generations. At the start of the tracking, it was found that a particular locus of one chromosome can be occupied by one of two sequences – Sequence A or Sequence B. These two sequences were observed to be non-coding.

The population size of 100 of the species has been roughly stable throughout the generations in island X. The number of individuals possessing Sequence A and Sequence B at the start of the tracking is shown in **Table 8.1**.

Sequence possessed	Number of individuals
Sequence A only	25
Sequences A and B	50
Sequence B only	25

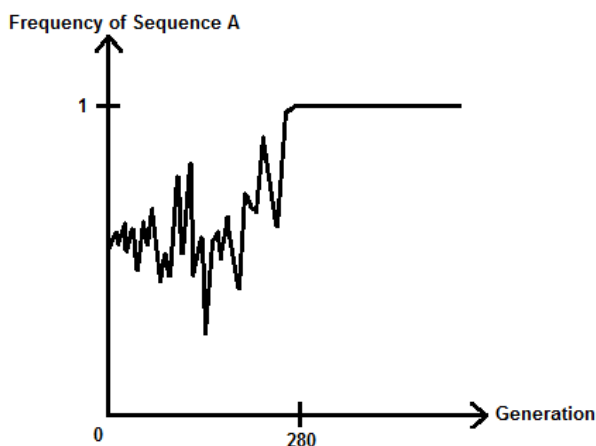
Table 8.1

- (a) In the space below, calculate the frequency of Sequence A in the population at the start of the tracking. [1]

$$[(25 \times 2) + 50] / (2 \times 100)$$

$$= 0.5 \text{ OR } 50\%$$

Fig. 8.2 shows the change in the frequency of Sequence A over many generations in the population on island X.

**Fig. 8.2**

- (b) With reference to **Fig. 8.2**, describe the change in the frequency of Sequence A. [2]
- Frequency of Sequence A** increased from 0.5 to **1** from generation 0 to **generation 280** and **remained at 1** after generation 280;
 - Ref. **random fluctuations** of frequency from **generation 0 to generation 280**;

(c) Provide an explanation for the change in the frequency of Sequence A over the generations. [3]

1. Sequence A (and Sequence B) is non-coding, thus, does not code for any protein and **not likely to affect the phenotype of the organism**;
2. Sequence A (and Sequence B) are thus **selectively neutral / does not confer a selective advantage or disadvantage** to the organism;
3. Random fluctuation and eventual fixation (Reaching a frequency of 1) of Sequence A is due to **genetic drift**;

The same species was also found to inhabit another remote island Y with a population size of 2000. The same two sequences of A and B were also found to occupy the same locus in this population. **Fig. 8.3** shows the change in the frequency of Sequence A over many generations in the population on island Y.

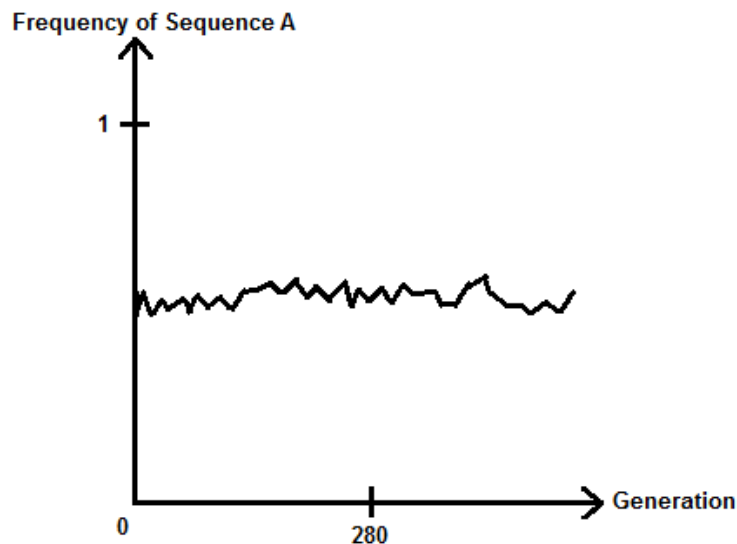


Fig 8.3

(d) Suggest and explain the differences in the frequency of the sequence A over many generations in the two populations on island X and Y. [2]

1. The **population on island Y** has a **much larger population size** compared to the one on island X;
2. The **effect of genetic drift** is **larger** on **smaller populations** (on island X) / OWTTE, thus, fixation (frequency of 1) is achieved faster at generation 280 on island X as compared to in island Y;

A related species Y was discovered. To determine if species X and Y are related, an investigator studied DNA sequences which codes for an important protein that performs the same function in both X and Y.

Sequence D (from species X) and Sequence E (from species Y) were identical except at two points.

- (e) Explain how Sequences D and E provide evidence to support the theory of evolution. [2]

1. The two sequences are very similar and show molecular homology/ are homologous sequences;
2. Hence it is most likely that the similarities were inherited from a common ancestor;
3. This is evidence for descent with modification as the two sequences show a modification process from an ancestral sequence via natural selection.

[Total: 10]

Section B

Question 9

- (a) Explain why and how eukaryotic pre-mRNA is processed. [6]

Max 4 for HOW/ WHY

1. **Capping**: addition of a modified guanine nucleotide at the 5' end of pre-mRNA;
2. **protects the mRNA** from **degradation by exonucleases** OR
3. functions as **part of a signal** for the **attachment of the ribosomes** for the **initiation of translation**;
4. **RNA splicing** - Introns are cut out and exons spliced/ joined together/ OWTTE;
5. Involves splicesomes, each consists of small nuclear ribonucleoproteins (snRNPs) with additional proteins;
6. A spliceosome **recognises splice sites**, found at the ends of an intron, **cleaves/cuts** both ends of the intron and **join/ ligate** the two exons that previously flanked the intron;
7. Produce a continuous coding sequence/mature mRNA, so that functional proteins can be formed;
8. **Alternative splicing** may occur, resulting in **different combinations of mature mRNA from a gene**, thus different polypeptides can be made from a single gene, increasing the number of proteins without the increase in number of genes;
9. **Polyadenylation**: addition of a poly(A) tail/ 50 to 250 adenine nucleotide at the 3' end of pre-mRNA;
10. **protects the mRNA** from **degradation by exonucleases/** longer poly(A) tail implies **longer half-life** of mRNA OR

11. facilitates the **export of the mRNA** out of the nucleus/ also helps **ribosomes** to attach to mRNA.

(b) Compare the two processes that have to take place for protein synthesis. [9]

Max 6 for differences and Max 3 for similarities

Similarities:

- (1) Both processes **require a template**.
- (2) Both processes are **catalyzed by enzymes**.
- (3) Both processes involve **complementary base-pairing**.
- (4) Both involve **condensation reactions with the removal of 1 water molecule per bond synthesized**.

Differences:

Feature	Transcription	Translation
Location	In <u>nucleus</u>	<u>ribosomes</u>
Product(s)	mRNA, rRNA and tRNA	Polypeptide/ protein
Template	<u>DNA</u> template	<u>Messenger RNA</u>
Enzyme	<u>RNA polymerase</u>	<u>Peptidyl transferase</u>
Bond between basic units.	RNA polymerase catalyses formation of <u>phosphodiester bond</u> between <u>ribonucleotides</u>	Peptidyl transferase catalyzes the formation of <u>peptide bond</u> between <u>amino acids</u> .
Monomers	free <u>ribonucleotides/ RNA nucleotides</u>	<u>amino acids</u> attached to tRNA
Reading of genetic message	RNA polymerase moves along DNA template in <u>3' to 5' direction</u>	Ribosome moves along mRNA in <u>5' to 3' direction</u>
Fate of product(s)	Products exit nucleus and migrate to the cytoplasm	Polypeptide chain/ protein remain in the cytoplasm or secreted out of the cell
Involvement of ribosome	not involved	<u>Ribosome</u> is the binding site for mRNA and amino-acyl tRNA complex.
Involvement of tRNA	not involved.	<u>tRNA</u> carries amino acids to ribosomes.

(c) Explain how genetic variation arises in sexually reproducing organisms. [5]

1. **Meiosis** is an important step for **sexual reproduction** as haploid gametes are produced and meiosis results in genetic variation;
2. Due to **crossing over of non-sister chromatids of homologous chromosomes**, at the **chiasmata**, during **prophase I**;
3. where **homologous regions/ regions containing same genes to be exchanged**;
4. creating **new combination of alleles**;
5. Due to **independent assortment of homologous chromosomes** during **metaphase I**,
6. whereby the **orientation** of homologous pair of chromosomes along the **equator/ metaphase plate** is **random/ independent of other bivalents**;
7. **Independent assortment** of **chromosomes/ pairs of chromatids** also occurs during **metaphase II**, significant due to **crossing over** which occurred during prophase I.
8. With independent assortment, this results large number of **different chromosome combinations in the gametes**. i.e. 2^n , where **n = number of homologous pairs of chromosomes or haploid number**;
9. In addition, during **fertilization, random fusion of gametes** occurs, resulting in numerous combinations of a zygote;

Maximum 1 mark on mutation;
10. **(Spontaneous) gene mutation**: change in **DNA sequence of a gene** / change in one or a few nucleotides, giving rise to **new alleles**;
11. **Chromosomal aberrations**: change in **structure** of chromosomes due to translocation / deletion / duplication of chromosomal fragment + elaboration;
12. **Chromosomal aberrations**: change in **number** of chromosome i.e aneuploidy / polyploidy – extra / lack of one chromosome or sets of chromosomes + elaboration;

Question 10

- (a) Outline the main features of oxidative phosphorylation and contrast this with photophosphorylation. [9]

Features of oxidative phosphorylation (Max 7)

1. **High energy electrons from NADH and FADH₂** are channelled to the **electron transport chain**;
2. **electrons** are passed along a **series of electron carriers of progressively lower energy levels** to oxygen, the final electron acceptor;
3. **energy released** from electron transfer is used to **actively transport/ pump protons from the matrix into the intermembrane space**;
4. This produces a **high concentration of protons** in the **intermembrane space**, setting up an electrochemical **proton gradient / proton motive force** across inner mitochondrial membrane;
5. As **H⁺ ions** **diffuse** from the **intermembrane space into the matrix** of the mitochondrion through **ATP synthase complex**;
6. the **energy released** is utilised by **ATP synthase** to catalyse the **synthesis of ATP** from **phosphorylation of ADP** with inorganic phosphate;
7. **Oxygen**, the **final electron acceptor**, combines with **protons** and electrons, and is **reduced to form water**, catalyzed by **cytochrome oxidase**;
8. **NADH and FADH₂** are **oxidised** to **NAD⁺ and FAD**, respectively/ **NAD⁺ and FAD** are **regenerated** at the end of oxidative phosphorylation

Max 4 for differences

	Photophosphorylation	Oxidative phosphorylation
Location	<u>Thylakoid membrane of chloroplast</u>	<u>Inner membrane of mitochondrion</u>
Electron donors	<u>Water</u> - Electron donor in non-cyclic pathway & <u>P700 specialised chlorophyll a</u> - Electron donor in cyclic pathway	Reduced coenzyme, NADH/FADH₂
Electron acceptors	<u>NADP</u> - Final electron acceptor in non-cyclic pathway & <u>P700 specialised chlorophyll a</u> - Final electron acceptor in cyclic pathway	Molecular <u>Oxygen</u>
Establishing proton gradient for the synthesis of ATP	H ⁺ ions are pumped inwards , from <u>stroma</u> across the thylakoid membrane, into the <u>thylakoid space</u> .	H ⁺ ions are pumped outwards , from <u>matrix</u> across the inner mitochondrial membrane, into the <u>intermembrane space</u> .
Source of energy for phosphorylation	<u>Light energy</u> from the sun	Oxidation of glucose / NADH / FADH
Involvement of light	<u>Light</u> is required for photolysis of water.	Light is not required.
Involvement of pigments	<u>Pigments</u> (chlorophylls a and b, carotene and xanthophyll) are used for trapping light energy.	Pigment is not required.
Products	<u>ATP, NADPH</u> + H ⁺ produced, <u>oxygen</u> as a by-product.	<u>ATP</u> and <u>water</u> produced; <u>NAD⁺</u> and <u>FAD</u> are regenerated.

(b) Compare the Krebs cycle and the Calvin cycle.

[6]

Max 4 for differences

Features	Calvin Cycle	Krebs Cycle
Location	Occurs in <u>stroma of chloroplast</u> .	Occurs in <u>matrix of mitochondrion</u> .
Cell type	Occurs in plant cells / algae / blue-green bacteria	Occurs in <u>all aerobically respiring</u> cells
Process	<u>Anabolic</u> reaction (formation of triose phosphate or starch)	<u>Catabolic</u> reaction (breakdown of acetyl-coA)
Electron or hydrogen carriers	<u>NADP+</u> (nicotinamide adenine dinucleonucleotide phosphate)	<u>NAD+ and FAD</u> (nicotinamide adenine dinucleonucleotide and flavin adenine dinucleotide)
Compound that is regenerated	<u>Ribulose biphosphate</u> is regenerated at the end of cycle.	<u>Oxaloacetate</u> is regenerated at the end of cycle.
Carbon dioxide	Carbon dioxide is <u>fixed</u> by ribulose biphosphate , catalysed by ribulose biphosphate carboxylase (Rubisco) .	Carbon dioxide released by oxidative decarboxylation .
ATP	ATP is used in the reduction of PGA to PGAL/ and the regeneration of RuBP.	ATP is synthesised by <u>substrate level phosphorylation</u> .
Redox reactions	- Reduced NADP are <u>used</u>; - To <u>reduce</u> phosphoglyceric acid (PGA) to triose phosphate	<u>Reduced NAD and FAD</u> are formed; - when NAD and FAD <u>oxidises</u> intermediates of the cycle
Role of O ₂	Does not require O₂	Occurs only when O₂ is present
Requirement of light	Is indirectly dependent on light as requires NADPH and ATP which were produced during the light dependent reactions .	Is not dependent on light
Enzymes involved	Carboxylases	Dehydrogenases, decarboxylases

Similarities of the Krebs Cycle and Calvin Cycle

Max 3 for similarities

1. Both involve **regeneration of the starting compound**, thus they are termed 'cycles' / cyclic processes;
 - In respiration, oxaloacetate is regenerated; in photosynthesis, RuBP is regenerated;
2. Both cycles involve **redox reactions**;
3. Both cycles utilize **electron / hydrogen carriers**;
 - Krebs cycle – NAD^+ & FAD^+ ; Calvin cycle – reduced NADP / NADPH

(c) Outline the process of fermentation in yeasts and explain its significance. [5]

1. **Pyruvate** is produced by **glycolysis** which undergoes **alcoholic fermentation in yeasts in absence of oxygen/anaerobic condition**;
2. **pyruvate** is first **decarboxylated** to form **acetaldehyde (or ethanal)** by **pyruvate decarboxylase** and **carbon dioxide is produced**;
3. **ethanal** acts as the **final electron and hydrogen acceptor**;
4. **ethanal** is then **reduced** to form **ethanol** by **NADH**, catalyzed by alcohol dehydrogenase.;
5. **Significance: to regenerate NAD^+** ;
6. for **glycolysis to proceed**, so there is **production of net 2 ATP per glucose during glycolysis** by substrate level phosphorylation under **anaerobic conditions/when oxygen is absent**.